

ActMix: Recovering Tabular Inductive Bias via Adaptive Activation Topology

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Abstract—Deep neural networks often underperform gradient-boosted decision trees on tabular data, partly due to a mismatch between smooth, homogeneous neural inductive biases and heterogeneous feature semantics. We propose ActMix, an adaptive activation layer that allows each neuron to learn a convex combination over a topologically diverse basis set (Sine, Tanh, ReLU, Identity) via learnable mixing logits. Temperature annealing and low-entropy regularization promote specialization, yielding sparse and interpretable activation topologies. We evaluate ActMix with 10-fold cross-validation on three UCI tabular benchmarks across various MLP architectures, reporting both predictive performance and fold-aggregated interpretability statistics. ActMix is competitive with strong neural baselines and approaches XGBoost performance on the MAGIC Gamma classification task while providing transparent, layer-wise activation usage patterns.

Index Terms—Tabular learning, adaptive activation functions, mixture of experts, neural architecture design, inductive bias, deep learning

I. INTRODUCTION

Despite remarkable progress in computer vision and natural language processing, deep neural networks remain inferior to gradient boosted decision trees, such as XGBoost, on tabular data [1], [2]. This persistent gap stems from fundamental differences in inductive biases: tree-based models excel at handling uninformative features, irregular decision boundaries, and heterogeneous data distributions, while neural networks impose smooth, rotationally invariant transformations that often misalign with tabular structure.

Traditional neural network architectures apply the same activation function uniformly across all neurons in a layer which is a design choice inherited from computer vision where spatial homogeneity is desirable. However, tabular datasets exhibit fundamentally different characteristics where features may require different nonlinear transformations, some may benefit from periodic encodings, and others from piecewise-linear approximations. This architectural rigidity limits the network’s ability to adapt to the heterogeneous nature of tabular data.

We introduce ActMix, an adaptive activation layer that enables neuron-level specialization by learning a convex combination over topologically diverse basis functions. Inspired by mixture-of-experts architectures [3], ActMix treats activation function selection as a learnable topology optimization, where

each neuron discovers its optimal functional form through gradient descent. Our basis set comprising ReLU (piecewise-linear), Tanh (sigmoidal), Sine (periodic), and Identity (linear) spans distinct topological spaces, enabling the network to recover inductive biases suited to tabular data.

To prevent trivial mean-field solutions where all neurons converge to identical mixtures, we employ temperature annealing and an entropy-minimization regularizer. The softmax temperature governing mixing coefficients is linearly annealed from 1.0 to 0.01, promoting initial exploration followed by crisp specialization. The entropy term encourages low-entropy mixing distributions, yielding neuron-level specialization and improving interpretability.

We evaluate ActMix on three UCI benchmark datasets (Adult Census Income, MAGIC Gamma Telescope, and Combined Cycle Power Plant) using 10-fold cross-validation and MLPs with one to three hidden layers. Our contributions are: (i) a neuron-level adaptive activation mechanism tailored to tabular feature heterogeneity; (ii) statistical comparisons across folds using non-parametric tests to quantify when improvements are reliable; (iii) an ablation study over depth and specialization regularization to isolate the effect of adaptive sharpening; and (iv) fold-aggregated interpretability analyses that characterize layer-wise topology evolution and neuron specialization.

II. RELATED WORK

A. Parametric and Static Improvements

The optimization of activation functions has historically focused on alleviating gradient flow issues within a fixed topological family. He et al. introduced PReLU [4] to mitigate the “dying ReLU” problem by learning a negative slope, while Ramachandran et al. utilized Neural Architecture Search (NAS) to discover Swish ($x \cdot \sigma(x)$) [5]. While these methods introduce learnable scalar parameters, they remain topologically static; a PReLU network is fundamentally piecewise linear and cannot approximate periodic or bounded manifolds efficiently.

B. Inductive Biases

Recent empirical benchmarks by Grinsztajn et al. [2] have systematically investigated why tree-based models continue to outperform deep learning on tabular data. Their findings suggest that the superior performance of methods like XGBoost stems from specific inductive biases such as robustness

to non-informative features, lack of rotational invariance, and the ability to model irregular decision boundaries. Standard neural networks, with their bias towards smooth, rotationally invariant functions, often struggle to approximate the sharp, non-continuous manifolds present in heterogeneous tabular datasets.

In addition to irregular boundaries, tabular data often contains periodic or high-frequency signals like seasonality and cyclical timestamps. Sitzmann et al. [6] demonstrated with SIREN that standard ReLU networks struggle to model such signals, requiring sinusoidal activations to capture fine details and derivatives. ActMix directly addresses these limitations by utilizing a topologically diverse activation space. By including both periodic (Sine) and sparse (ReLU) bases, it allows the network to recover these necessary inductive biases at the neuron level.

C. Mixture-Based Architectures

The concept of adaptive activation mixtures has been previously explored in Computer Vision by Manessi and Rozza [7]. They proposed learning combinations of standard functions, such as ReLU, Tanh, Identity, within Convolutional Neural Networks, allowing the network to dynamically approximate complex non-linearities for image classification tasks. More recently, Li et al. [8] proposed CombU, which employs linear combinations of activations to fit mathematical expressions in symbolic regression tasks. While these works establish the viability of mixture-based activations, Manessi et al. focus on spatially structured image data, and CombU targets formula discovery. Neither approach addresses the unique challenges of tabular data, such as feature heterogeneity, where each column represents a distinct physical quantity with unique statistical properties.

D. Deep Learning Architectures

Gorishniy et al. [9] conducted a rigorous evaluation of deep learning architectures for heterogeneous tabular data, highlighting the lack of established benchmarks in the field. Their study introduced the FT-Transformer, a simple adaptation of the Transformer architecture that embeds categorical and numerical features into a unified space, demonstrating performance often competitive with Gradient Boosted Decision Trees, such as XGBoost. While their work focuses on architectural modifications to handle heterogeneity, it lacks the analysis and effect of activation functions.

Arik and Pfister [10] proposed TabNet, another prominent architecture that employs sequential attention mechanisms to mimic the feature selection capabilities of decision trees within a differentiable framework. TabNet dynamically selects relevant features at each decision step, offering interpretability and efficient learning. However, like FT-Transformer, TabNet’s innovation lies in its macroscopic architecture rather than the microscopic behavior of individual processing units.

E. Learnable Activation Functions

Optimization of activation functions has evolved beyond static choices, acknowledging that deciding on the best activa-

tion is non-trivial and often depends on the architecture, hyperparameters, and the dataset itself. Early work by Agostinelli et al. [11] introduced Adaptive Piecewise Linear (APL) units, allowing networks to learn the slopes and hinge points of activation functions during training. While this marked a shift towards adaptive non-linearities, APL units are restricted to piecewise linear shapes. This limitation restrains the network’s capacity to model smooth or periodic phenomena, which are common in tabular datasets.

To address smoothness, Molina et al. [12] introduced Padé Activation Units (PAU), utilizing rational functions (ratios of polynomials) to approximate flexible non-linearities end-to-end. While PAUs demonstrate that deviating from standard activations like ReLU can improve performance by learning a more suitable shape, they typically learn a single global function or set of parameters for the entire layer. However, rational functions are inherently smooth and constrained by their polynomial order, limiting their ability to model sharp transitions or high-frequency periodic signals efficiently. Moreover, learning a single flexible shape per layer fails to address the feature heterogeneity of tabular data, where individual neurons may require topologically distinct activations to capture specific physical properties.

III. METHOD

A. ActMix: Adaptive Activation Topology

We introduce ActMix, an adaptive activation layer designed to recover tabular inductive biases in deep neural networks. Drawing inspiration from Mixture-of-Experts [3], ActMix learns a neuron-specific convex combination over a topologically diverse basis set Φ . For a pre-activation input $z \in \mathbb{R}$, the activation output is:

$$\text{ActMix}(z) = \sum_{k=1}^K \alpha_k \phi_k(z) \quad (1)$$

where α_k represents the mixing coefficient for the k -th basis function ϕ_k . We employ a basis set $\Phi = \{\text{ReLU}, \text{Tanh}, \text{Sine}, \text{Identity}\}$ to span piecewise-linear, sigmoidal, periodic, and linear functional spaces. For the periodic component, we use $\sin(\omega_0 z)$ with a fixed frequency $\omega_0 = 1.0$ (lower than the $\omega_0 = 30$ used in SIREN [6], suited to tabular feature scales).

The mixing coefficients α are generated via a temperature-scaled softmax applied to learnable logits $\mathbf{l} \in \mathbb{R}^K$:

$$\alpha_k = \frac{\exp(l_k/\tau)}{\sum_{j=1}^K \exp(l_j/\tau)} \quad (2)$$

The mixing logits $\mathbf{l}$ are initialized to zero, establishing a uniform prior over basis functions at the start of training.

To evaluate the contribution of dynamic specialization, we investigate two distinct variants of ActMix. The first variant employs temperature annealing and entropy regularization to encourage specialization. In this configuration, the temperature τ is linearly annealed from 1.0 to 0.01 over the first 100 epochs, promoting initial exploration followed by convergence toward a crisp selection. Additionally, we minimize the entropy of the mixing distribution $H(\alpha)$ via a regularization

term $\lambda \cdot H(\alpha)$, where λ is warmed up from 0.0 to 0.01 over 100 epochs. The second variant serves as an ablation study to isolate the effect of adaptive sharpening; it operates without these regularizers (fixed $\tau = 1.0$ and $\lambda = 0.0$ throughout training). This forces the model to rely solely on gradient descent to learn the optimal mixture without explicit sparsity pressure.

B. Datasets

We validate ActMix on three UCI tabular datasets representing diverse domains. The Adult Census dataset presents a heterogeneous classification task predicting income levels, containing 45,222 samples after removing rows with missing values and 104 features after one-hot encoding [13]. The MAGIC Gamma Telescope dataset is a binary classification benchmark comprising 19,020 samples and 10 features [14]. Finally, the Combined Cycle Power Plant dataset offers a regression task predicting net hourly electrical energy output, with 9,568 samples and 4 features [15]. Data preprocessing includes standard scaling for numerical features and one-hot encoding for categorical variables. For each fold, the held-out 10% serves as the validation set for early stopping; no additional inner split is employed.

C. Baselines

We compare ActMix against standard MLPs equipped with fixed activation functions: ReLU, GELU [16], and PReLU [4]. Additionally, we benchmark against Gradient Boosted Trees (XGBoost) [1], the dominant paradigm in tabular learning [9]. XGBoost models are configured with 500 estimators, a learning rate of 0.05, and a maximum depth of 6.

D. Training Protocol

All deep learning models are trained using the AdamW optimizer [17] with a learning rate of 10^{-3} and weight decay of 10^{-4} . We employ a cosine annealing scheduler with a minimum learning rate factor of 0.01. Training proceeds for 200 epochs with a patience of 20 epochs for early stopping. To ensure stability, we apply gradient clipping with a maximum norm of 0.5.

We utilize a batch size of 128 and a dropout rate of 0.1 across all MLP variants. To analyze the impact of depth, we evaluate one, two, and three hidden-layer MLPs, each with 128 neurons per layer. Classification tasks optimize the Cross-Entropy loss, while the regression task minimizes the Mean Squared Error (MSE). All experiments are seeded with 1192 to ensure reproducibility. All experiments were conducted on 2× NVIDIA T4 GPUs with 30GB of RAM.

E. Evaluation

We employ 10-fold stratified cross-validation for classification tasks and 10-fold cross-validation for regression. This evaluation protocol reduces variance arising from data splitting.

IV. RESULTS

We present a comprehensive evaluation of ActMix across classification and regression tasks, followed by an ablation study and a detailed analysis of the learned activation topologies.

A. Comparative Performance

We compare ActMix against standard MLP baselines equipped with ReLU, GELU, and PReLU activation functions. Additionally, we include XGBoost as a reference baseline for tabular performance. Table I summarizes the results across all three datasets.

On the MAGIC Gamma classification task, ActMix achieves an accuracy of 0.8858 ± 0.0046 , performing comparably to GELU (0.8838 ± 0.0037) and significantly outperforming ReLU (0.8807 ± 0.0026). This result supports our hypothesis that adaptive activation functions can recover inductive biases critical for certain tabular domains. On the heterogeneous Adult Census dataset, ActMix achieves an accuracy of 0.8565 ± 0.0044 , surpassing both ReLU (0.8549) and PReLU (0.8545) baselines in mean performance. For the Power Plant regression task, ActMix achieves an R^2 of 0.9337 ± 0.0046 and an RMSE of 4.381 ± 0.180 , demonstrating strong predictive capability competitive with the best neural baselines (GELU).

B. Statistical Significance

To rigorously validate our findings, we performed statistical significance testing across the 10 cross-validation folds. We employed the Friedman test to detect overall differences among neural models [18], followed by Holm-adjusted Wilcoxon signed-rank tests for pairwise comparisons. For the MAGIC Gamma dataset, the Friedman test indicated significant differences ($p = 0.0037$). Pairwise comparisons revealed that ActMix significantly outperformed ReLU ($p = 0.012$), though the difference with GELU was not statistically significant ($p = 0.39$). In the Power Plant regression task, the Friedman test confirmed significance ($p < 0.001$), and ActMix showed statistically significant improvement over both ReLU ($p = 0.018$) and PReLU ($p = 0.018$) baselines. For the Adult Census dataset, the Friedman test indicated significant overall differences ($p = 0.014$); however, subsequent pairwise comparisons did not yield statistically significant differences after Holm adjustment (all $p > 0.05$).

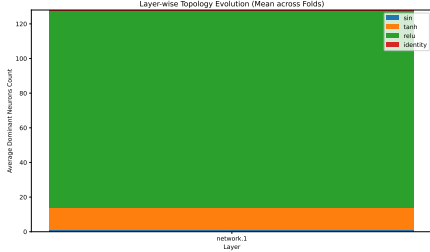
C. Ablation Study: Depth and Specialization

To isolate the impact of our temperature annealing and entropy regularization strategy, as well as network depth, we compare the full ActMix formulation against a variant without these regularizers (Table II).

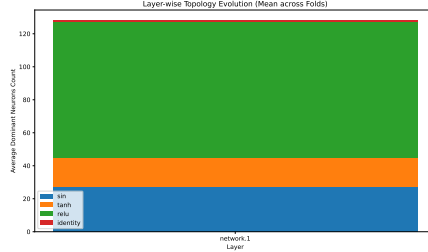
The results demonstrate that the “adaptive sharpening” mechanism (regularization) yields consistent improvements on Magic Gamma and Power Plant across depths. On Adult Census, the effect is mixed within small margins, suggesting that regularization is more critical for tasks requiring sharp decision boundaries. For instance, on MAGIC Gamma with a 3-layer network, the full ActMix model achieves 0.8858

TABLE I
MAIN PERFORMANCE COMPARISON. WE REPORT THE BEST-PERFORMING DEPTH (1, 2, OR 3 LAYERS) FOR EACH MODEL/DATASET. ACTMIX USES THE REGULARIZED VARIANT. BEST NEURAL MODEL IN **BOLD**. XGBOOST IS PROVIDED AS A REFERENCE BASELINE.

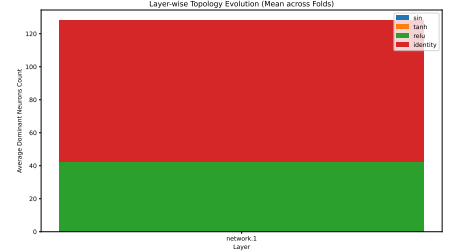
Model	Adult Census		Magic Gamma		Power Plant	
	Accuracy $\uparrow$	F1 $\uparrow$	Accuracy $\uparrow$	F1 $\uparrow$	R^2 $\uparrow$	RMSE $\downarrow$
ActMix (Ours)	0.8565 $\pm$ 0.0044	0.7970 $\pm$ 0.0069	0.8858 $\pm$ 0.0046	0.8705 $\pm$ 0.0052	0.9337 $\pm$ 0.0046	4.381 $\pm$ 0.180
GELU	0.8560 $\pm$ 0.0045	0.7956 $\pm$ 0.0080	0.8838 $\pm$ 0.0037	0.8678 $\pm$ 0.0043	0.9351 $\pm$ 0.0042	4.334 $\pm$ 0.165
ReLU	0.8549 $\pm$ 0.0041	0.7929 $\pm$ 0.0072	0.8807 $\pm$ 0.0026	0.8644 $\pm$ 0.0028	0.9316 $\pm$ 0.0057	4.450 $\pm$ 0.212
PReLU	0.8545 $\pm$ 0.0037	0.7925 $\pm$ 0.0054	0.8853 $\pm$ 0.0059	0.8695 $\pm$ 0.0070	0.9319 $\pm$ 0.0052	4.439 $\pm$ 0.199
XGBoost (Ref.)	0.8705 $\pm$ 0.0053	0.8164 $\pm$ 0.0080	0.8859 $\pm$ 0.0033	0.8710 $\pm$ 0.0037	0.9668 $\pm$ 0.0048	3.098 $\pm$ 0.239



(a) Adult Census



(b) Magic Gamma



(c) Power Plant

TABLE II
ABLATION STUDY: IMPACT OF DEPTH AND REGULARIZATION (ADAPTIVE SHARPENING).

Depth	Reg.	Adult Accuracy	Magic Accuracy	Power Plant R^2
1 Layer	Yes	0.8547	0.8749	0.9302
	No	0.8553	0.8722	0.9292
2 Layers	Yes	0.8565	0.8815	0.9337
	No	0.8554	0.8799	0.9309
3 Layers	Yes	0.8564	0.8858	0.9274
	No	0.8574	0.8837	0.9243

accuracy compared to 0.8837 for the unregularized variant, confirming that forcing the network to make crisp decisions among basis functions enhances generalization in suitable domains.

D. Interpretability and Learned Topology

A central claim of ActMix is its ability to learn interpretable, neuron-specific activation functions. We analyze the learned mixing coefficients to understand the topological structure of the trained networks. The presented figures represent fold-aggregated results (mean $\pm$ standard deviation) to ensure robustness.

Figure 2 further illustrates the distribution of the maximum mixing coefficient ($\max_k \alpha_k$) across all neurons in the MAGIC Gamma network. The distribution is strongly bimodal, with a significant mass near 1.0, confirming that the entropy regularization successfully drives neurons to specialize.

Finally, Figure 3 visualizes representative effective activation functions learned by specific neurons. We see diverse

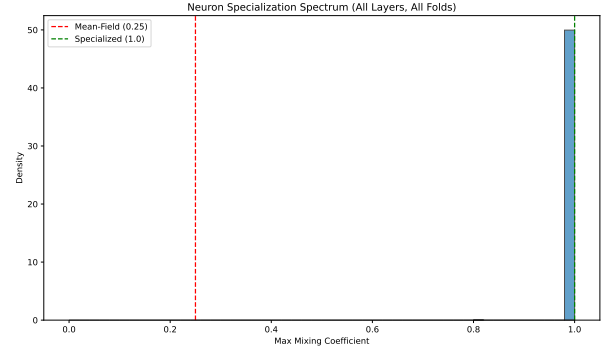


Fig. 2. Neuron Specialization Spectrum. The histogram of maximum mixing coefficients shows a bimodal distribution, indicating that most neurons specialize in a single basis function (values near 1.0) rather than using a dense mixture.

behaviors. For example, some neurons converge to standard ReLUs, others to identity mappings, and some to periodic functions.

V. DISCUSSION

A. Performance Analysis

Our results demonstrate that ActMix can narrow the performance gap between deep neural networks and gradient-boosted decision trees on specific tabular benchmarks. On the MAGIC Gamma dataset, ActMix achieves an accuracy of 0.8858, closely matching the performance of XGBoost (0.8859). This suggests that improving neuron-level nonlinearity can mitigate part of the tabular inductive-bias mismatch identified in prior benchmarks [2]. While specialized tabular architectures like FT-Transformer [9] and TabNet [10] address data heterogeneity through attention mechanisms and sparse feature selection,

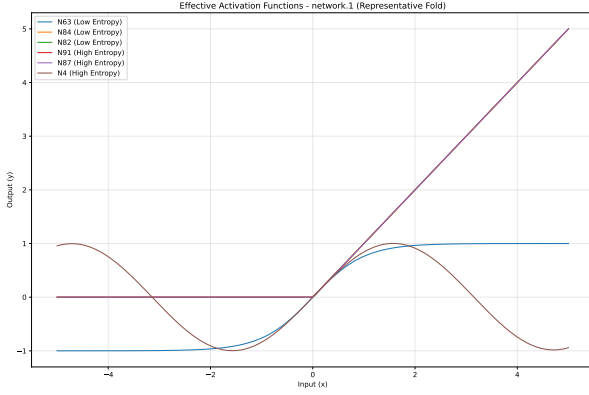


Fig. 3. Representative Learned Effective Activations. Visualization of the composite activation functions $\sum \alpha_k \phi_k(z)$ for selected neurons in the first hidden layer, showing the diversity of learned non-linearities.

ActMix targets the fundamental processing unit. By significantly outperforming standard ReLU and PReLU baselines on the Power Plant regression task ($p < 0.05$, Holm-adjusted Wilcoxon), we show that recovering tabular inductive bias does not necessarily require complex macro-architectures, but can be achieved by relaxing the homogeneity constraint of the activation function.

B. Interpretability and Topology

The learned activation topologies provide insight into how ActMix adapts to data heterogeneity. As illustrated in Fig. 2, the distribution of mixing coefficients is strongly bimodal. This indicates that neurons effectively “specialize” into specific functional forms, a mechanism conceptually aligned with Mixture-of-Experts (MoE) architectures [3], [19]. However, unlike the sparse gating in MoEs which routes samples to experts, ActMix adapts the functional topology itself.

This neuron-level specialization distinguishes ActMix from prior work on learned activations. Approaches like Padé Activation Units (PAU) [12] or Adaptive Piecewise Linear (APL) units [11] typically learn a single, global shape for a layer. Similarly, search-based methods that discovered Swish [5] or architectural components like Maxout [20] aim to find a single optimal function. In contrast, ActMix acknowledges that tabular datasets contain semantically distinct features that may benefit from diverse transformations. By allowing simultaneous existence of periodic (Sine), sigmoidal (Tanh), and piecewise-linear (ReLU) units within the same layer (Fig. 1), ActMix captures the “irregularity” inductive bias [2] that previously gave tree ensembles their edge. This aligns with Manessi and Rozza’s findings in computer vision [7], but extends the principle to the heterogeneous nature of tabular feature spaces.

C. Ablation and Regularization

The ablation study (Table II) confirms the critical role of our regularization strategy. The performance gain observed with temperature annealing and entropy minimization indicates that “adaptive sharpening” is essential for effective specialization.

Without these mechanisms, the network tends to learn soft, entangled mixtures that fail to capture the sharp decision boundaries characteristic of tabular data. The regularization forces the network to commit to specific functional forms, thereby reducing the hypothesis space and improving generalization.

D. Limitations

Despite these gains, ActMix introduces computational overhead during both training and inference, as it requires evaluating multiple basis functions for every neuron. While the number of parameters increases only marginally (due to the mixing weights), the floating-point operations (FLOPs) scale linearly with the size of the basis set. Additionally, while ActMix outperforms standard MLPs, the performance gap with XGBoost on high-cardinality categorical datasets suggests that activation topology alone may not fully address the challenges of discrete feature representation, which may require specialized embedding strategies.

E. Future Work

Future research should explore learnable parameters within the basis functions themselves, such as the frequency ω_0 in the Sine component, to allow for multi-scale periodic feature extraction. Furthermore, integrating ActMix into Transformer-based tabular architectures, such as FT-Transformer [9], could combine the benefits of attention mechanisms with adaptive non-linearities. Finally, investigating the connection between the learned sparse topologies and Neural Architecture Search (NAS) could yield more efficient, pruned inference models that retain the benefits of adaptive activation without the runtime cost of the full basis set.

VI. CONCLUSION

We presented ActMix, a novel neural architecture that recovers tabular inductive bias through adaptive activation topology. By enabling neurons to learn convex combinations of topologically distinct basis functions, ActMix achieves performance competitive with XGBoost on the MAGIC Gamma classification task and significantly outperforms ReLU/PReLU baselines on the Power Plant regression benchmark. Our analysis reveals that this performance stems from the network’s ability to evolve from linear to specialized non-linear processing layers. These findings suggest that relaxing the homogeneity constraint of activation functions is a principled path toward closing the gap between neural networks and tree-based models on tabular data, though XGBoost remains dominant on high-cardinality tasks.

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